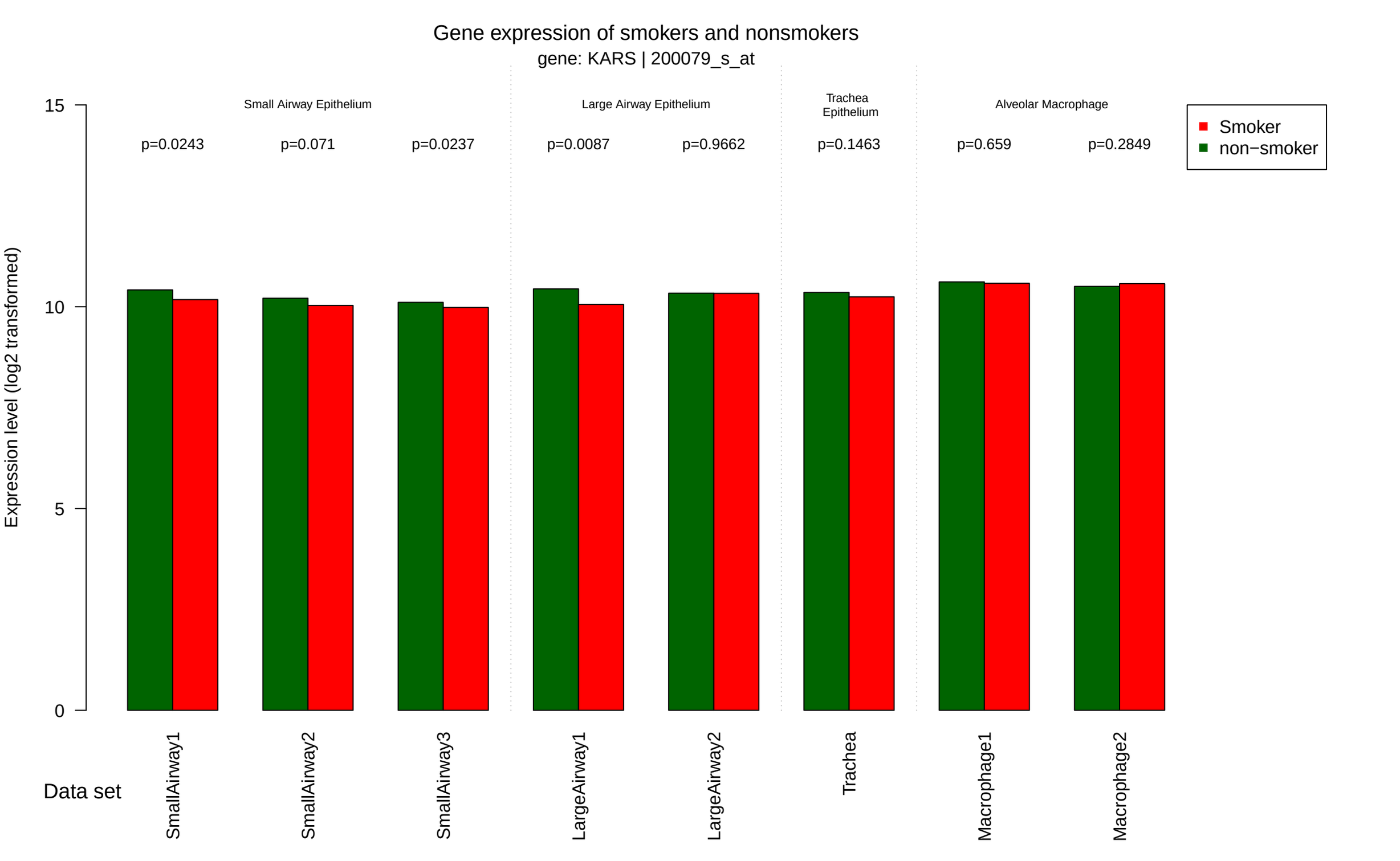
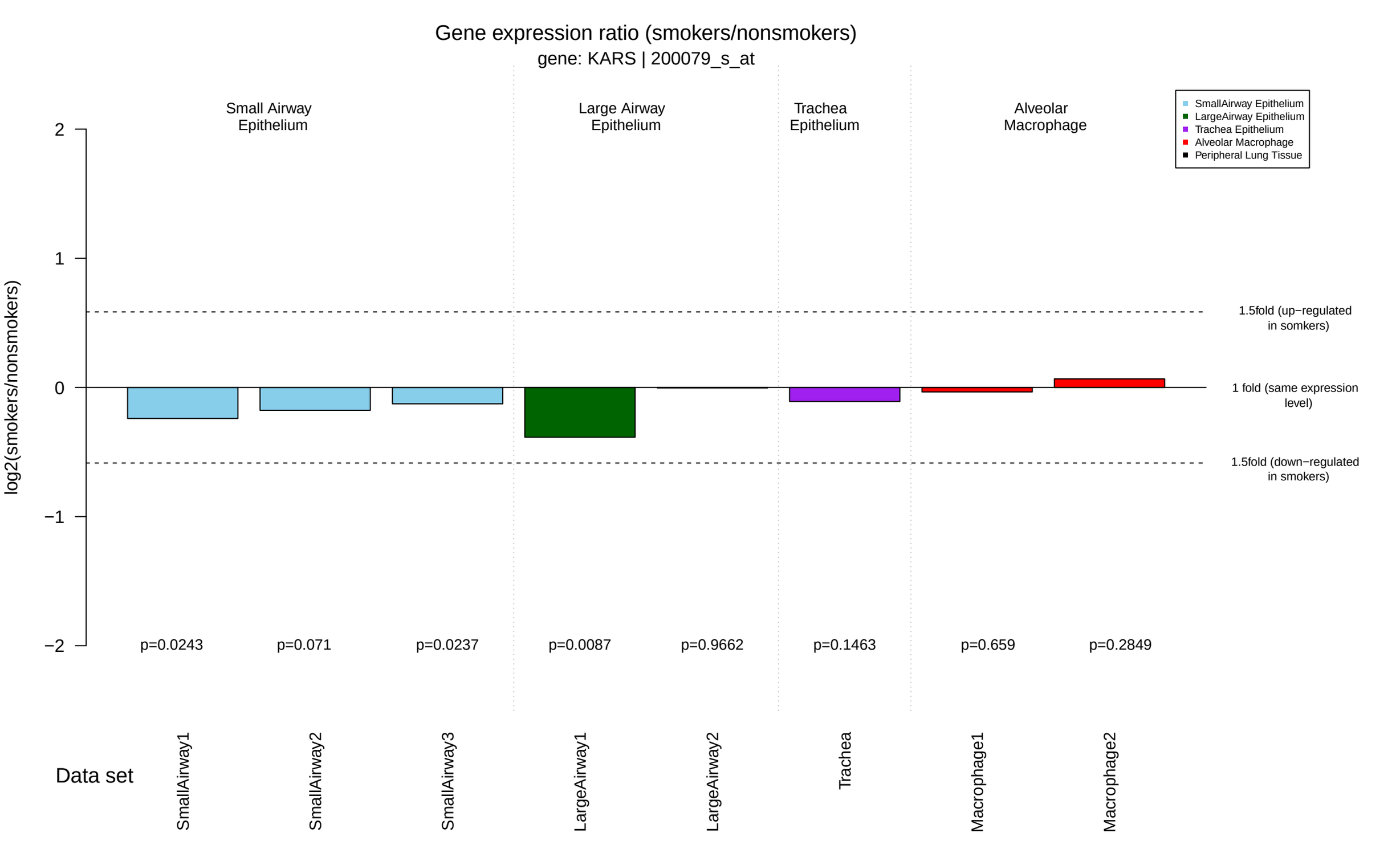
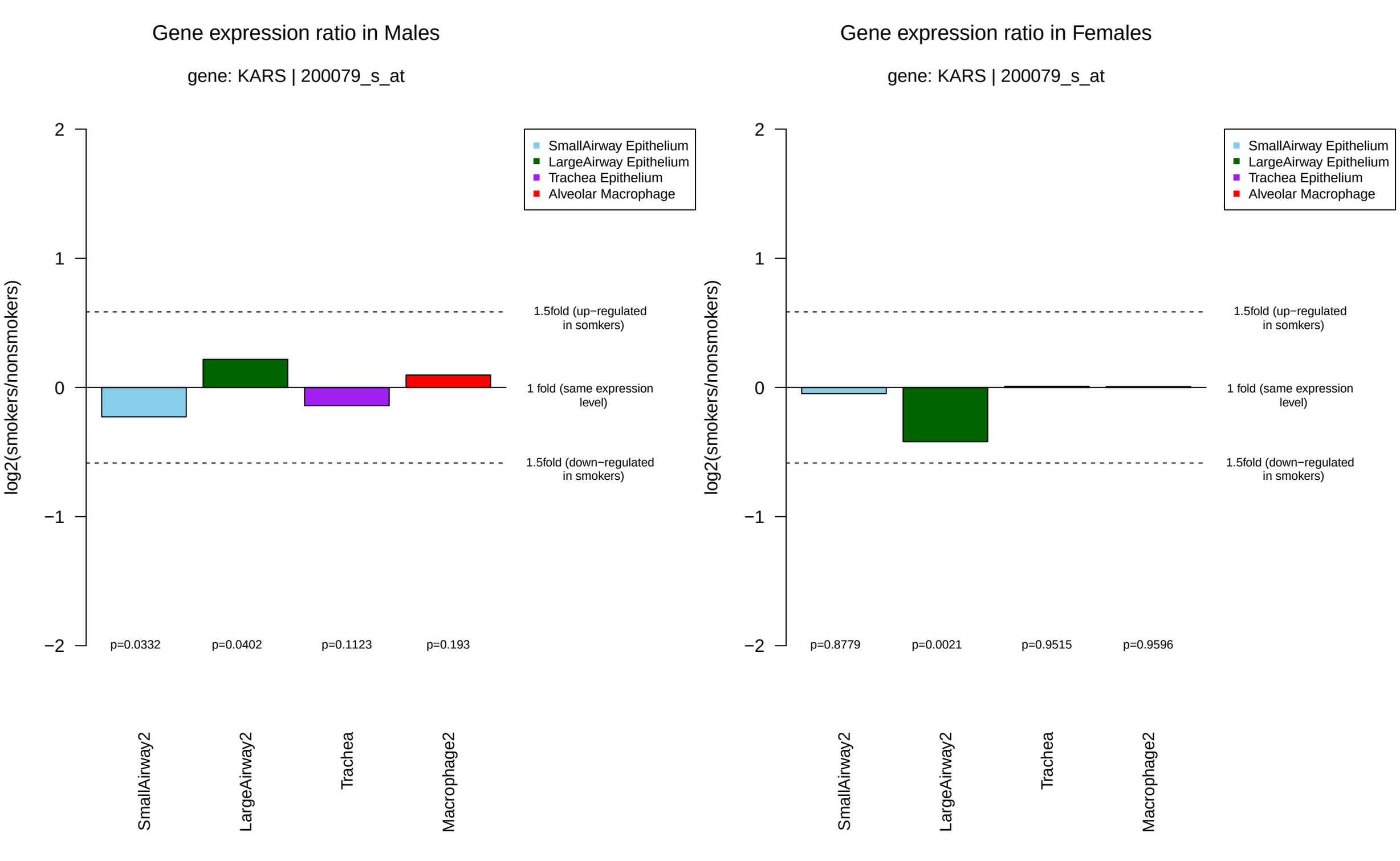
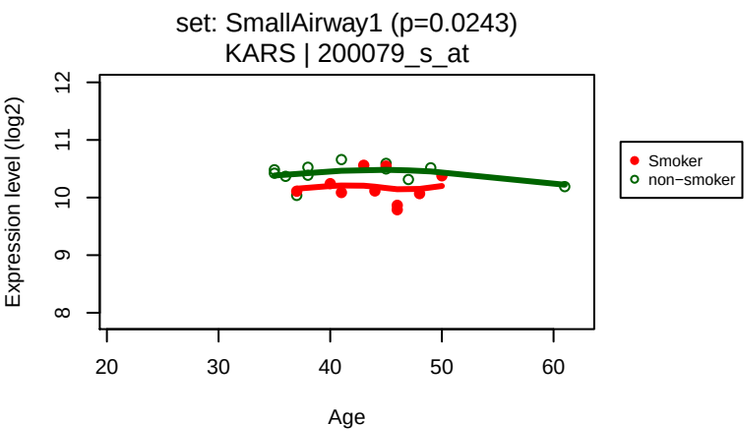
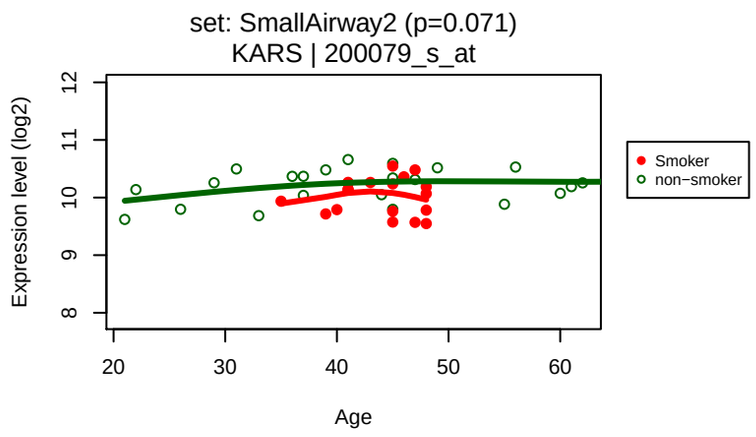
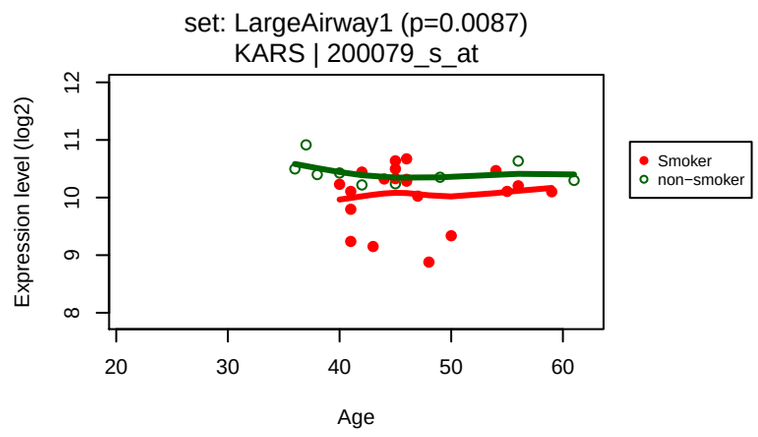

Gene expression of smokers and nonsmokers
gene: KARS | 200079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0243
p=0.071
p=0.0237
p=0.0087
p=0.9662
p=0.1463
p=0.659
p=0.2849
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

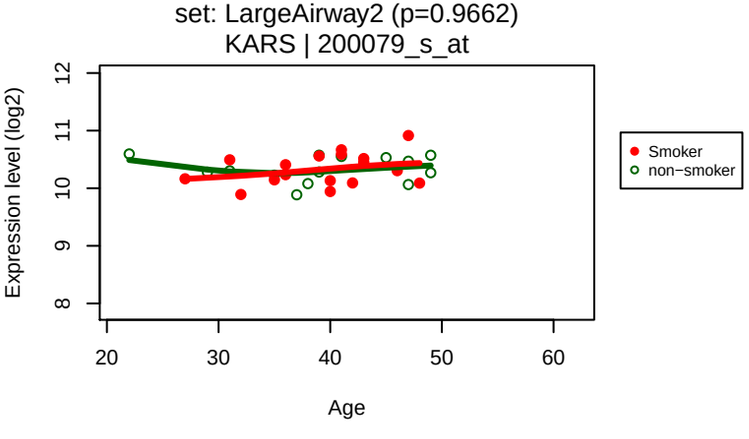

Gene expression ratio (smokers/nonsmokers)
gene: KARS | 200079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0243
p=0.071
p=0.0237
p=0.0087
p=0.9662
p=0.1463
p=0.659
p=0.2849
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

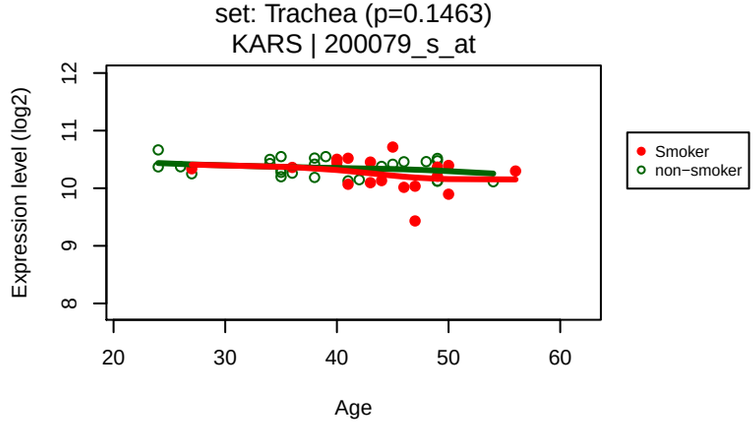

Gene expression ratio in Males
gene: KARS | 200079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0332
p=0.0402
p=0.1123
p=0.193
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KARS | 200079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8779
p=0.0021
p=0.9515
p=0.9596
SmallAirway2
LargeAirway2
Trachea
Macrophage2

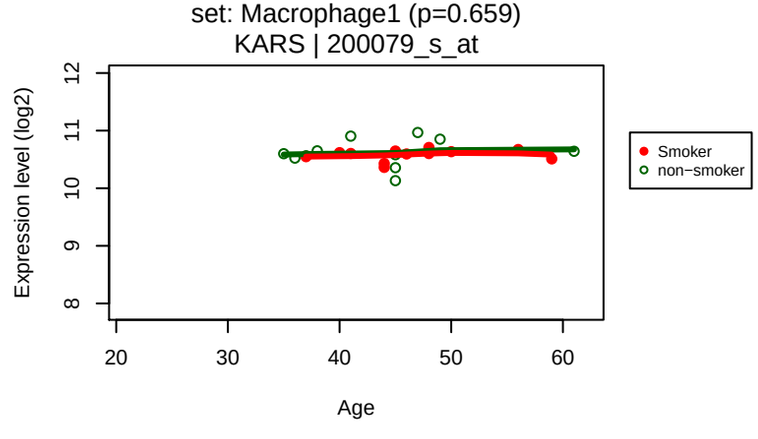

set: SmallAirway1 (p=0.0243)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker

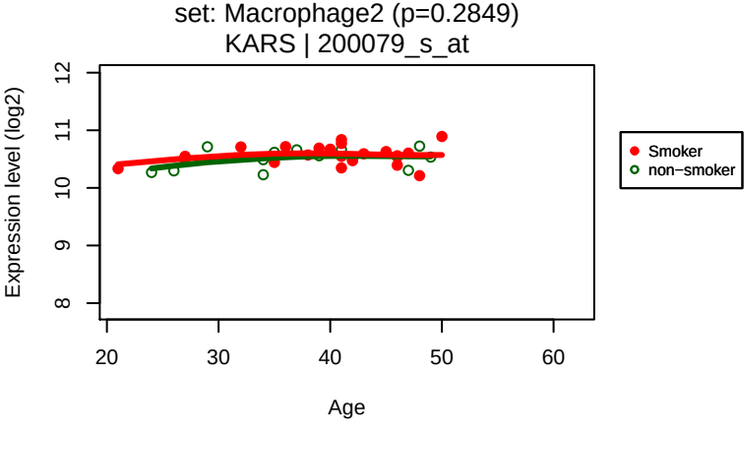

set: SmallAirway2 (p=0.071)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker

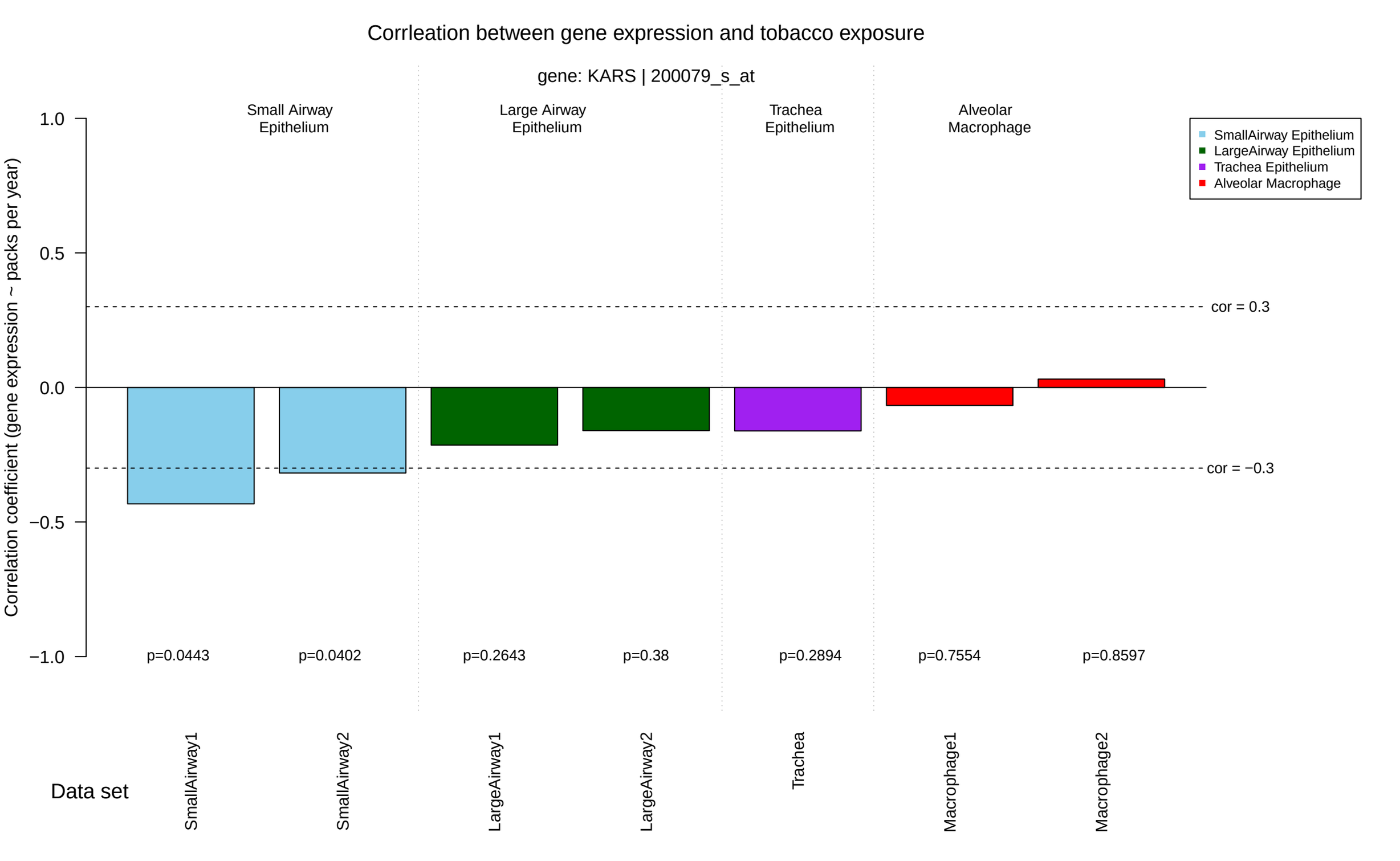

set: LargeAirway1 (p=0.0087)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9662)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1463)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.659)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2849)
KARS | 200079_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KARS | 200079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0443
p=0.0402
p=0.2643
p=0.38
p=0.2894
p=0.7554
p=0.8597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2